



ELTE | FACULTY OF  
INFORMATICS



**SAP**  
**University**  
**Alliances**

# **Proceedings of the**

# **1st SAP UA Community Conference**

## **Central and Eastern Europe**

28-29 Oct. 2024, Budapest, Hungary

Sponsored by SAP University Alliance Program

Editor: Prof. Tamás Orosz

General Chair: Prof. Udo Bub

Program Committee Chair: Prof. Tamás Orosz

Published by Eötvös Loránd University Faculty of Informatics  
ISBN 978-963-489-736-1

2024, Budapest

# Proprietary vs. Open-Source ERP: The Case of SAP and Odoo

Ferenc Erdős  
Department of Informatics  
Széchenyi István University  
Győr, Hungary  
erdosf@sze.hu

Barnabas Isoke Kasangaki  
Information Technology Section  
Kibaale District Local Government  
Kibaale, Uganda  
ekbarnabas26@gmail.com

László Koloszar  
Alexandre Lamfalussy Faculty of  
Economics Institute of Business Studies  
University of Sopron  
Sopron, Hungary  
koloszar.laszlo@uni-sopron.hu

**Abstract**—Getting Enterprise Resource Planning (ERP) systems into place is one of the key components that businesses need to drive operations, transactions, reporting, and process automation. While open-source solutions have achieved remarkable progress in nearly every software product category for businesses, ERP has remained mostly proprietary in real installations, and open-source ERP systems account for a relatively small percentage of the ERP market overall. This study compares many aspects of the industry leaders in proprietary ERP (SAP) with open-source ERP (Odoo) to examine the reasons for the dominance of proprietary ERP.

**Keywords**— ERP, open-source, SAP, Odoo

## I. INTRODUCTION

Any business or organization may succeed with the help of information technology. Putting Enterprise Resource Planning (ERP) systems into place is one of the key components that businesses need to drive operations, transactions, reporting, and process automation. Businesses often produce more at the lowest feasible cost and respond quickly to the constantly shifting needs and demands of both internal and external consumers.

An open-source is a software where the source code is available for other developers to use, modify and give enhancements. The product-associated revenue streams are shifted toward auxiliary products such as support, training, documentation, and others [16]. Unlike proprietary software, open-source software is available for use freely and modified under terms defined by the software license [17].

Open-source solutions have made significant progress across various software sectors within enterprises, but the implementation of ERP systems has remained largely proprietary, with open-source ERP solutions occupying only a small portion of the overall ERP market [8]. This article explores the factors contributing to the dominance of proprietary ERP systems and compares the characteristics of market leaders in proprietary ERP (SAP) and open-source ERP (Odoo).

## II. DEFINITION OF ERP

From a process perspective, an ERP system is a comprehensive suite of IT applications that facilitates and supports enterprise operations [1]. ERP systems are software platforms that span the entire enterprise, utilizing an integrated database [2]. The support provided can encompass functions or processes. ERP software streamlines operations, data, and information flow throughout a company by combining essential business activities into a single system. [7]. This is critical for efficient resource management, improved productivity, and well-informed decision-making [7]. An ERP serves as a single source of truth, such as SAP S/4HANA's

HANA database, for all enterprise operations [1]. Users of an ERP within an enterprise are diverse, including end users, auditors, consultants, employees, and developers. ERPs have evolved over the years in response to the evolving business and market dynamics in their implementation [2]. ERPs should be installed within a technical environment that supports and aligns with overall corporate operations and strategy in order to guarantee that they function as viable platforms for integration and value generation [2]. Enterprises are faced with the decision of choosing between proprietary and open-source ERP systems. However, deciding between open-source and proprietary options is not straightforward [3].

## III. RESEARCH OBJECTIVES

There is a definite need to compare Odoo and SAP as they are two well-known ERP solutions on the market, each with unique features and capabilities catered to various company needs. SAP is proprietary and Odoo is open-source. A comparison of the two leaders on either model is a clear representation of the two different ERP software models. Khiller argues that open source has many developers and programmers who are least bothered by the idea of proprietary software, but on the other hand, the open-source poses a threat to the proprietary ERP industry [3]. This paper thus explores a comparative analysis of the proprietary and open-source software taking the case of SAP and Odoo to give some insight and guide to enterprises in their ERP implementations.

The research focused on the following objectives:

- To establish a comparative analysis of the proprietary and open-source ERP systems based on two market leaders.
- To gain insight into the implementation of proprietary and open-source ERP
- To examine the comparative implementation of SAP and Odoo
- To generate recommendations for better implementation of ERP in different environments as value addition.

## IV. PROPRIETARY VS. OPEN-SOURCE

An ERP system is considered proprietary when its internal code is not openly accessible to users and restrictions are enforced through the End User License Agreement (EULA). The EULA legally binds users to the terms and conditions governing the implementation of the ERP system [3]. Only the individual or organization who created the ERP software can make modifications to the source code [3]. The source code of the software is exclusive since its intellectual property rights are held only by the publisher or owner of the program. SAP is a prime example of a proprietary ERP system.

Open-source ERP systems are developed and tested through collaborative efforts, allowing anyone with the necessary expertise to access, modify, and distribute the source code [3]. The concept of open-source software originated from the GNU project in 1983 [3]. This collaborative approach aims to create software for all, enabling the sharing and editing of the source code with no restrictions for any purpose. This fosters communities of open-source software developers, users, and programmers. Open-source ERP systems like Odoo operate under this model.

There is a claim that in recent years, Odoo has gained significant traction in the ERP industry and is often referred to as the 'SAP killer' by many (kerning code,2023) [7]. This assertion serves as a primary motivation for this research. The popularity of proprietary ERP systems stems from their well-documented guidelines and established practices. SAP, for instance, incorporates thousands of best practices into its applications [1]. Open-source ERP systems like Odoo, ERPNext, Dolibarr, ERP5, ADempiere, and Compiere are evolving [1]. However, the global market share of open-source ERP systems has remained at around 3% of the entire ERP market [9]. Odoo accounts for over 50% of the small market share held by open-source ERP systems.

## V. SAP

SAP, established in 1972 in Germany by five former IBM engineers, is a world-renowned provider of software for business process management (SAP, 2024) [4]. The company has evolved into a global enterprise with its headquarters in Walldorf, Germany, and over 105,000 employees worldwide (SAP Global Communications 2024). Its ERP software has set the global standard, and now, with SAP S/4HANA, the company is taking ERP to the next level by utilizing in-memory computing to enable cutting-edge technologies like machine learning and artificial intelligence (AI) and manage enormous volumes of data.

SAP S/4HANA, the company's in-house ERP system, replaces the previous, process-driven platform by integrating apps that connect all facets of a business into one intelligent suite on a completely digital platform. With over 100 solutions covering all business operations and the largest cloud portfolio among providers, SAP counted over 230 million cloud users in 2024 [4]. The SAP ERP system provides various business functions with a single source of truth through centralized data management, facilitating better management by providing workers from many departments with simple access to real-time data throughout the company, even for complicated business processes. As a result, companies may increase productivity, streamline processes, improve customer satisfaction, increase operational efficiency, and eventually increase profitability.

Any business may use SAP to collect and handle data on a single platform for anything from manufacturing and customer satisfaction to the sourcing of raw materials. SAP helps businesses analyze and efficiently develop the complete value chain and may be used "on-premise" at the user's site or via the cloud.

SAP HANA is SAP's built an in-memory, column-oriented, relational database management system known as HANA, or "High-Performance Analytic Appliance" [15]. Using the Intel Xeon CPU, the SAP HANA appliance software functions as a flexible, memory-based platform that

integrates different SAP software components designed for hardware provided by leading SAP technology partners such as Cisco, Dell, IBM, HP, and Fujitsu. The SAP HANA database, data and lifecycle management apps, support for several industry standard interfaces, and the SAP HANA studio—a user-friendly tool for information modeling and administration—are all included in this appliance's integrated SAP software components. The newest data management platform from SAP is built on top of the SAP HANA database.

The main goal of the SAP HANA database is to provide a strong and flexible platform for different types of queries, such as analytical and transactional ones, with the same data model in an extremely scalable execution environment [12]. Compared to all other ERP data management architectures, SAP HANA offers better and more efficient data access and administration than the traditional SQL database structures used in Odoo. SAP HANA is geared for column-based storage and supports both row-based and column-based storage [15]. Figure 1 shows the overview of the HANA database layered architecture.

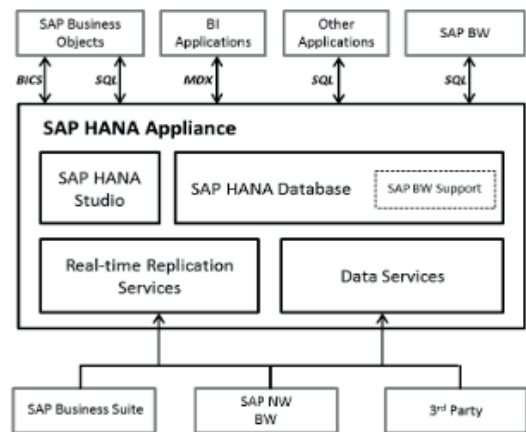


Fig. 1. Overview of the HANA database layered architecture (Source: [12])

The SAP HANA database differs from traditional database systems as it aims to act as a versatile platform supporting various domain-specific languages [12]. The system's core concept is based on a flexible data flow model, where query expressions and scripts are linked to the model, while different physical operators use the same table layer interface for complete record life cycle management. To ensure transactional consistency, logging and data area are utilized to maintain a persistent storage copy of the main memory database [12].

SAP S/4HANA Cloud is specifically designed to cater to the needs of rapidly growing medium to large enterprises across all industries, offering extensive functionality and a high degree of adaptability [13]. The data integration feature in SAP S/4HANA allows seamless integration with SAP and other applications and business processes, surpassing other ERP solutions [14]. This is evidence of SAP's significant impact on the ERP industry. The architecture of SAP S/4HANA facilitates parallelization and integration with open-source technologies like Google Cloud [14]. HANA generates a considerable amount of cache in the database, leading to rapid processing [15]. The efficiency and superior data management capabilities of SAP HANA position it well above open-source ERPs, such as Odoo, in implementation.

The SAP Fiori UI has been created with tiles to provide a user experience similar to that of consumer applications in the corporate software environment. Its main goal is to develop user-friendly, role-specific apps that can be used on different devices.

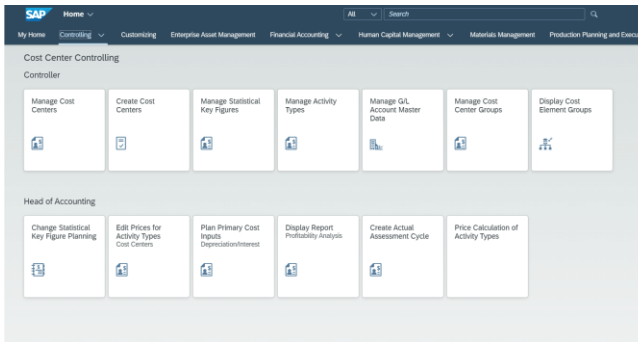


Fig. 1. Example of the SAP Fiori UI

## VI. ODOO

Odoo, a collection of open-source business applications, can be tailored to suit any business: CRM, eCommerce, accounting, inventory, point of sale, project management, and more[5]. Its distinguishing feature is its simultaneous ease of use and full integration. Originating in Belgium, its primary aim was to offer businesses a flexible and affordable ERP platform that wouldn't drain their finances.

Odoo ERP is a comprehensive software utilized across the entire company to manage business operations and drive automation. According to Odoo, business software should address complex requirements without being convoluted [7]. Odoo seeks to deliver software that is user-friendly, feature-rich, tightly integrated, and effortless to upgrade, ensuring seamless operation for all businesses and users. As the most widely used open source business software globally, Odoo is free to download and modify [5].

A smooth and welcoming user experience has been constructed to facilitate seamless user adoption. A large community of developers actively maintains Odoo to meet changing customer requirements and introduce new, innovative applications. Odoo's flexibility allows for the addition of apps as the company grows. It offers an adaptable and user-friendly interface designed to streamline business management, providing easy access to various modules and features.



Fig. 2. Example of the Odoo's UI

In the past 18 years, Odoo has expanded to become a prominent provider of management software, with large offices and partners worldwide, serving numerous customers.

Odoo is committed to providing top-notch software for businesses to efficiently manage their operations, enabling them to concentrate on running their business [6].

## VII. DETAILED COMPARISON OF SAP AND ODOO

In the following a detailed analysis will be provided comparing the two ERPs in terms of different dimensions. We selected some aspects which can be relevant for different companies. Suitability means the business size environment that the ERP can best be deployed to get the best out of it. The development dimension is connected to the program code aspect of the systems. The settings for operating the application are known as hosting preferences. Either the supplier handles all maintenance and availability (as in the case of software as a service models) and the firm pays license costs, or you may download the program for free onto your own server and handle all upkeep and maintenance yourself. Customization means the tailoring the ERP source code to the needs of the business' operations and processes. Data migration refers to moving data and its structure from the current systems to the ERP solution such that its compatible and usable in the ERP. Debugging refers to correcting and troubleshooting issues, errors and logical mistakes that may arise in the ERP source code. Adaptability means having the ERP modified to accommodate new business demands and changing processes and operations.

TABLE I. COMPARISON OF SAP S/4HANA AND ODOO

| <i>Dimension</i>             | <i>SAP S/4HANA</i>                                                                                                      | <i>ODOO</i>                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Suitability                  | Large multinational enterprises with big volume of transactions                                                         | Best for Small and Medium sized enterprises                          |
| Development                  | Proprietary written in higher level languages like C++, Java and ABAP                                                   | Open source written in python                                        |
| Hosting preferences          | Licence fees which exempts server maintenance                                                                           | No licence fees can even download and install on own servers         |
| Customization & adaptability | Sets requirements for customization as dogma<br><br>Has more complex structure to adjust the source code to new demands | More flexible to customize<br>Very flexible to adhere to new demands |
| Data migration               | Has backward compatibility for data transfer with ultimately no coding                                                  | Has scripts to ease data transfer                                    |
| Debugging                    | Rely on SAP partner or SAP technical support                                                                            | Relies on community of developers                                    |

It is clear from the detailed comparison table above that with Odoo, companies have control over the ERP code in addition to their main business, but with SAP, they pay and can focus on their core business. A vote akin to the superior ERP does not exist. Every ERP has its strengths and weaknesses. During choosing over the ERP systems, the size of the business, the amount of data handled, the budget, and internal expertise should all be taken into account.

Odoo works well for small and medium-sized businesses with uncomplicated business procedures. Odoo modules' capability of modification, adaptability, and flexibility in the face of a tight budget allows them to handle nearly any typical procedure of work.

SAP S/4HANA is an advanced platform created for large enterprises with intricate workflows. S/4HANA has been developed to handle and store enormous volumes of data. This makes it appropriate for big multinational businesses in a highly competitive market. The company can focus on its primary duties. The ERP obligations such as availability and maintenance are left to SAP at a fee for instance in cloud deployments.

### VIII. ERP COST CONSIDERATIONS

Considering the costs associated with open source and proprietary ERP implementations, as outlined in the case, is essential. These costs include acquisition, customization, testing, upgrade, conversion, personal development, unforeseen expenses, time, troubleshooting, and risk. Making well-informed decisions about ERP selection and implementation will lead to success once these costs are thoroughly evaluated and analyzed. When comparing SAP HANA and Odoo, the decision is based on the particular requirements of the company. While larger organizations tend to appreciate SAP HANA's extensive functionality, smaller firms frequently pick Odoo's user-friendly interface and cost-effectiveness. Additionally, the decision is influenced by the budget allocated for ERP implementation and the cost-benefit analysis.

### IX. CONCLUSIONS

The comparison of the Odoo and SAP ERP systems ultimately shows that the optimal option is contingent upon the particular business requirements, budget, and size. Larger businesses may benefit from SAP's powerful and feature-rich ERP solution, which delivers extensive capability across a variety of business activities. For companies needing worldwide assistance and strong ERP capabilities, its all-inclusive approach is perfect. Conversely, smaller firms may tailor an ERP solution to their own operational requirements with greater flexibility and cost-effectiveness thanks to Odoo's open-source nature and modular architecture. Ultimately, both Odoo and SAP ERP solutions have their strengths and areas of specialization. Businesses embarking on the ERP selection journey should consider their current operations, growth plans, and the total cost of ownership when deciding between Odoo and SAP. By aligning ERP capabilities with business needs, companies can ensure they invest in a system that not only meets their requirements today but can also scale and evolve alongside their future aspirations. The choice between Odoo and SAP is a significant one, paving the way for enhanced efficiency, productivity, and competitive advantage in the dynamic business landscape of 2024

In the end, each of the SAP ERP and Odoo systems has advantages and areas of expertise. When choosing between Odoo and SAP, businesses should take into account their growth goals, present operations, and total cost of ownership. Businesses can make sure they invest in a system that not only fits their needs now but can also grow and change to meet their goals in the future by matching ERP capabilities with company demands. The decision made between Odoo and SAP will have a big impact on productivity, efficiency, and competitive advantage in the ever-changing corporate world.

### X. SUMMARY

SAP ERP is a robust, scalable solution with comprehensive functionality, but it comes with a higher cost and complexity. Odoo ERP offers a more affordable, flexible, and user-friendly alternative, but with several limitations in industry-specific solutions and integration capabilities. A large company with complex operations and industry-specific needs should choose SAP S/4HANA with scalability and robustness. For SMEs, Odoo ERP can be a cost-effective solution, with huge flexibility. The choice depends on the requirements of the business, its size, and its budget.

### References

- [1] Marianne Radford, (2012) Modern ERP: Select, Implement, and Use Today's Advanced Business Systems,
- [2] Shadrack Katuu (2020) Enterprise Resource Planning: Past, Present, and Future, New Review of Information Networking, 25:1, 37-46, DOI:10.1080/13614576.2020.1742770
- [3] Sagar Khillar (2024) Difference Between Open Source and Proprietary Software. <http://www.differencebetween.net/technology/difference-between-open-source-and-proprietary-software/> accessed 05/08/2024
- [4] SAP(2024) <https://www.sap.com/about/what-is-sap.html>
- [5] Odoo(2024) <https://www.odoo.com/page/about-us>
- [6] Fabien Pinckaers (Odoo), 2020 est le manager de l'année 2020 - RTBF Actus <https://www.rtf.be/article/fabien-pinckaers-odoo-est-le-manager-de-l-annee-2020-10666813>
- [7] Odoo vs SAP ERP: A Comprehensive Comparison for 2024. <https://www.kerningcode.com/blog/odoo-comparison-4/odoo-vs-sap-erp-a-comprehensive-comparison-for-2024-23>
- [8] Mordor Intelligence.(2022).Global open Source Enterprise resource planning market report (2022-2027).The USA.
- [9] Odoo(2024) <https://www.odoo.com/forum/help-1/the-openERP-s-total-market-share-in-the-world-53497>.
- [10] Sage (UK) Limited. (2022). Conducting a cost-benefit analysis for ERP. North Park. Newcastle [https://www.sage.com/~media/group/files/global%20campaign/sage\\_cost\\_benefit\\_analysis\\_ERP.pdf](https://www.sage.com/~media/group/files/global%20campaign/sage_cost_benefit_analysis_ERP.pdf).
- [11] SAP Global Communications: SAP Corporate Fact Sheet (July 22, 2024) <https://www.sap.com/documents/2017/04/4666ecdd-b67c-0010-82c7-eda71af511fa.html>
- [12] Färber, Franz & May, Norman & Lehner, Wolfgang & Große, Philipp & Müller, Ingo & Rauhe, Hannes & Dees, Jonathan. (2012). The SAP HANA database - An architecture overview. IEEE Data Eng. Bull.. 35. 28-33.
- [13] Capterra (2024) Odoo vs SAP S/4HANA Cloud: Which is a better fit? <https://www.capterra.com/compare/135618-152293/Odoo-vs-SAP-S-4HANA>
- [14] Abdulrahman Nahhas, Christian Haertel, Christian Daase, Matthias Volk, Achim Ramesohl, Heiko Steigerwald, Alexander Zeier, Klaus Turowski (2023) On the Integration of Google Cloud and SAP HANA for Adaptive Supply Chain in Retailing. Procedia Computer Science, Volume 217, Pages 1857-1866, ISSN 1877-0509, <https://doi.org/10.1016/j.procs.2022.12.386>. (<https://www.sciencedirect.com/science/article/pii/S1877050922024711>)
- [15] Chetan Khemchand Bhojwani, Pradip S. Ingle (2019) SAP HANA High-Performance Analytical Appliance. Journal of Emerging Technologies and Innovative Research (JETIR) [www.jetir.org](http://www.jetir.org)
- [16] Kowanda, D., Firdaus, M., Bismark, R., & Pasaribu, F. (2015). Opportunity of Free Open Source ERP System as a Competitive Advantage for Small and Medium Enterprise. The 1st UICRIC 2015 Conference, DOI: 10.13140/RG.2.1.1473.1281.
- [17] Magnusson, A. (2016). A Framework for Selecting an ERP Open Source System: A Case Study. Master's Thesis. Department Of Computer Science Faculty of Engineering. Lund University.